1 Validation of the relation between residual correlations and selection bias

1.1 Motivation – inconsistency

Why a larger residual correlation, for example $\rho_{\ln Y_{\rm cyl},\ln \Sigma}$, does not by itself guarantee a larger selection bias in the stacked profile. The key point is that the selection bias is not determined only by a correlation coefficient. It also depends on how extreme the selected objects are in the residual distribution of the selection proxy, and on how the target signal is stacked.

We let

$$m_i \equiv \ln M_{500{\rm c},i}, \tag{1}$$

and let the selection proxy be

$$S \in \{N_{\rm cyl},\, Y_{\rm cyl}\}. \tag{2}$$

The target signal profile is

$$T(r_p) \in \{\Sigma(r_p),\, y(r_p)\}. \tag{3}$$

We work in log space:

$$s_i \equiv \ln S_i, \qquad t_i(r_p) \equiv \ln T_i(r_p). \tag{4}$$

For any log observable $x_i = \ln X_i$, define the KLLR mean relation at fixed halo mass as

$$\mu_X(m_i) \equiv \langle \ln X \mid \ln M_{500{\rm c}} = m_i \rangle\,. \tag{5}$$

The raw log residual is

$$\delta \ln X_i \equiv \ln X_i - \mu_X(m_i). \tag{6}$$

For a profile quantity,

$$\delta \ln T_i(r_p) \equiv \ln T_i(r_p) - \mu_T(m_i, r_p). \tag{7}$$

The KLLR function `kllr_model.residuals()` returns the standardized residual,

$$\epsilon_X \equiv \frac{\delta \ln X}{\sigma_X(m)}, \tag{8}$$

where $\sigma_X(m)$ is the local scatter around the KLLR mean relation. Therefore, the correlation coefficients measured are

$$\rho_{S,T}(r_p) = {\rm Corr}\left[\epsilon_S,\, \epsilon_T(r_p)\right], \tag{9}$$

not directly ${\rm Corr}[\delta \ln S, \delta \ln T]$, unless the scatter is nearly mass independent (we knew this is not true).

1.2 Selection strength

For a selected sample A_S, chosen using proxy S, we can define the selection strength

$$\eta_S \equiv \langle \epsilon_S \rangle_{A_S} - \langle \epsilon_S \rangle_{\rm parent}\,. \tag{10}$$

This quantity measures how far into the high-residual tail the selection goes, in units of the proxy scatter.

If the standardized residuals (ϵ_S, ϵ_T) were jointly Gaussian, and if the selection depended only on ϵ_S,

then the expected shift in the standardized target residual would be

$$\Delta\epsilon_T^{\rm pred}(r_p) = \rho_{S,T}(r_p)\,\eta_S. \tag{11}$$

The directly measured value is

$$\Delta\epsilon_T^{\rm obs}(r_p) = \langle\epsilon_T(r_p)\rangle_{A_S} - \langle\epsilon_T(r_p)\rangle_{\rm parent}\,. \tag{12}$$

Thus, for the same target profile, proxy $N_{\rm cyl}$ can only give a larger residual response than $Y_{\rm cyl}$ if

$$\rho_{N,T}(r_p)\,\eta_N > \rho_{Y,T}(r_p)\,\eta_Y. \tag{13}$$

Equivalently, when all quantities are positive, we have

$$\frac{\eta_N}{\eta_Y} > \frac{\rho_{Y,T}}{\rho_{N,T}}. \tag{14}$$

Therefore, a larger correlation coefficient alone is not sufficient to predict the larger selection bias.

To convert approximately from standardized residual shifts to raw log-residual shifts, we use

$$\Delta_{\rm res}^{\ln T,{\rm pred}}(r_p) \simeq \rho_{S,T}(r_p)\,\eta_S\,\langle\sigma_T(m_i,r_p)\rangle_{A_S}\,. \tag{15}$$

This approximation is exact **only if the scatter is independent of mass, or if the selected and parent samples have the same relevant scatter distribution.**

We ca directly measure the raw log-residual shift via

$$\Delta_{\rm res}^{\ln T,{\rm obs}}(r_p) = \langle\delta\ln T(r_p)\rangle_{A_S} - \langle\delta\ln T(r_p)\rangle_{\rm parent}\,. \tag{16}$$

1.3 Total log shift, mass-trend shift, and stacked ratio

The raw target signal can be decomposed as

$$\ln T_i(r_p) = \mu_T(m_i,r_p) + \delta\ln T_i(r_p). \tag{17}$$

Therefore, the selected-sample mean log shift is

$$\Delta_{\rm tot}^{\ln T}(r_p) \equiv \langle\ln T(r_p)\rangle_{A_S} - \langle\ln T(r_p)\rangle_{\rm parent} \tag{18}$$

$$= \underbrace{\Big[\langle\mu_T(m,r_p)\rangle_{A_S} - \langle\mu_T(m,r_p)\rangle_{\rm parent}\Big]}_{\Delta_{\rm mass}^{\ln T}(r_p)} + \underbrace{\Big[\langle\delta\ln T(r_p)\rangle_{A_S} - \langle\delta\ln T(r_p)\rangle_{\rm parent}\Big]}_{\Delta_{\rm res}^{\ln T}(r_p)}. \tag{19}$$

The first term is the **mass-trend shift**. It measures the change in the mean signal caused by selecting a different halo-mass distribution. The second term is the **residual shift** at fixed mass.

The stacked signal ratio is measured in linear space:

$$R_T(r_p) \equiv \frac{\langle T(r_p)\rangle_{A_S}}{\langle T(r_p)\rangle_{\rm parent}} = \frac{\langle\exp[\ln T(r_p)]\rangle_{A_S}}{\langle\exp[\ln T(r_p)]\rangle_{\rm parent}}. \tag{20}$$

In general,

$$R_T(r_p) \neq \exp\left[\Delta_{\rm tot}^{\ln T}(r_p)\right], \tag{21}$$

because $\langle e^x \rangle \neq e^{\langle x \rangle}$. Therefore, the linear stack ratio can be larger than the mean-log shift, especially for broad or non-Gaussian signal distributions.

1.4 Controlled validation test

I perform a controlled test using the same parent halo sample for both proxies. The richness catalogue gives $N_{\rm cyl}$ for 73325 matched halos with $q_{\rm proj} = 180\,h^{-1}$cMpc, $n_g = 10^{-2}$, $M_z = 20$, and $\chi^2 = 6$. Although $Y_{\rm cyl}$, Σ, and y are available for all 83731 halos above the mass cut, here we restrict the $Y_{\rm cyl}$ analysis to the same 73325 richness-matched halos. This removes the parent-sample difference between the $N_{\rm cyl}$ and $Y_{\rm cyl}$ tests. (In `Y_correlation_add_data.ipynb`, `richness_correlation_add_data.ipynb`, and Y/$N_{\rm cyl}$-selected selection bias results in `XXX_selection_XXX.ipynb`, the selected halos numbers are different. It is unfair to compare Y/richness selected sample with different halo numbers, and this may also cause an inconsistency you observed. I also note that in the `Y_selection_plot_add_data.ipynb`, we use the $Y_{\rm cyl}$ with noise and the $Y_{\rm sph}$ from Flamingo halo catalog to select $\sim$ 100-500 halos. But for the correlation coefficient calculation, I use the $Y_{\rm cyl}$ without noise and all 83731 halos.)

For each proxy, I select the top 10% of halos ranked by standardized proxy residual,

$$\mathrm{rank}_N = \epsilon_N, \qquad \mathrm{rank}_Y = \epsilon_Y. \tag{22}$$

This is different from selecting by $N_{\rm cyl}$ or $Y_{\rm cyl}$. Ranking by standardized residual isolates the fixed-mass residual-correlation effect. Ranking by the proxy value would also change the halo mass distribution and would mix the mass-trend term with the residual term.

I run two versions of the test:

1. **No mass-bin selection:** select the top 10 per cent over the full parent sample.
2. **Mass-bin selection:** divide the parent sample into mass bins and select the top 10 per cent within each bin.

1.5 Results

Figure 1 and Fig. 2 show the response test without and with mass-bin selection (two results are nearly same). The controlled test gives the same qualitative result for both $\Sigma(r_p)$ and $y(r_p)$:

$$\rho_{Y,T}(r_p) > \rho_{N,T}(r_p), \tag{23}$$

and

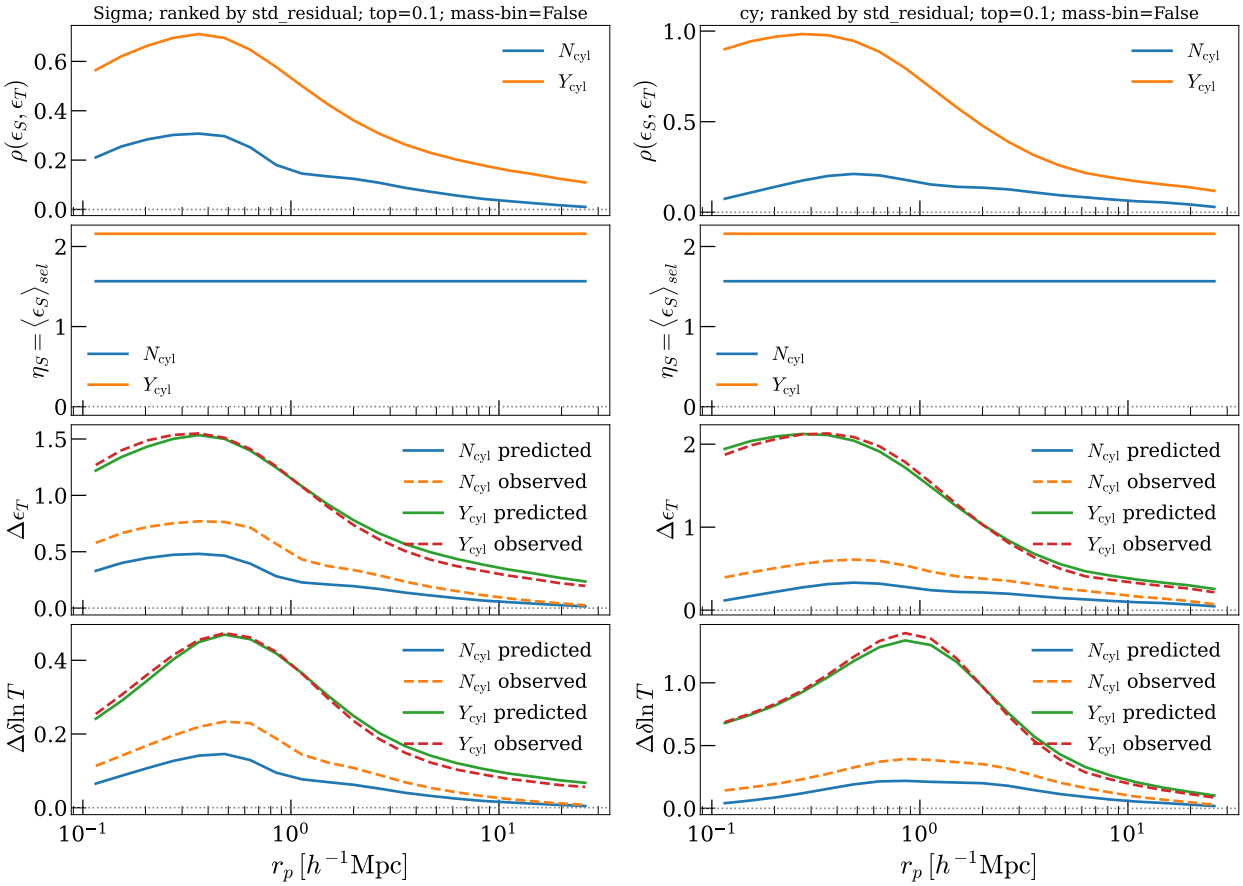
$$\eta_Y > \eta_N. \tag{24}$$

Thus the condition in Eq. (14) is not satisfied for richness to have the larger fixed-mass residual response. The $Y_{\rm cyl}$-selected sample has both the stronger residual correlation with the target profile and the more extreme selected proxy-residual tail. Therefore it gives the larger residual response in this controlled test.

Without mass-bin selection, the selected samples have slightly lower mean halo mass than the parent sample:

$$\Delta\langle \ln M \rangle_N \simeq -0.035, \qquad \Delta\langle \ln M \rangle_Y \simeq -0.030. \tag{25}$$

This produces a small negative mass-trend contribution. **However, the residual shift dominates the total log shift. See Fig. 3 and 4.**



(a) $\Sigma(r_p)$

(b) Compton-y

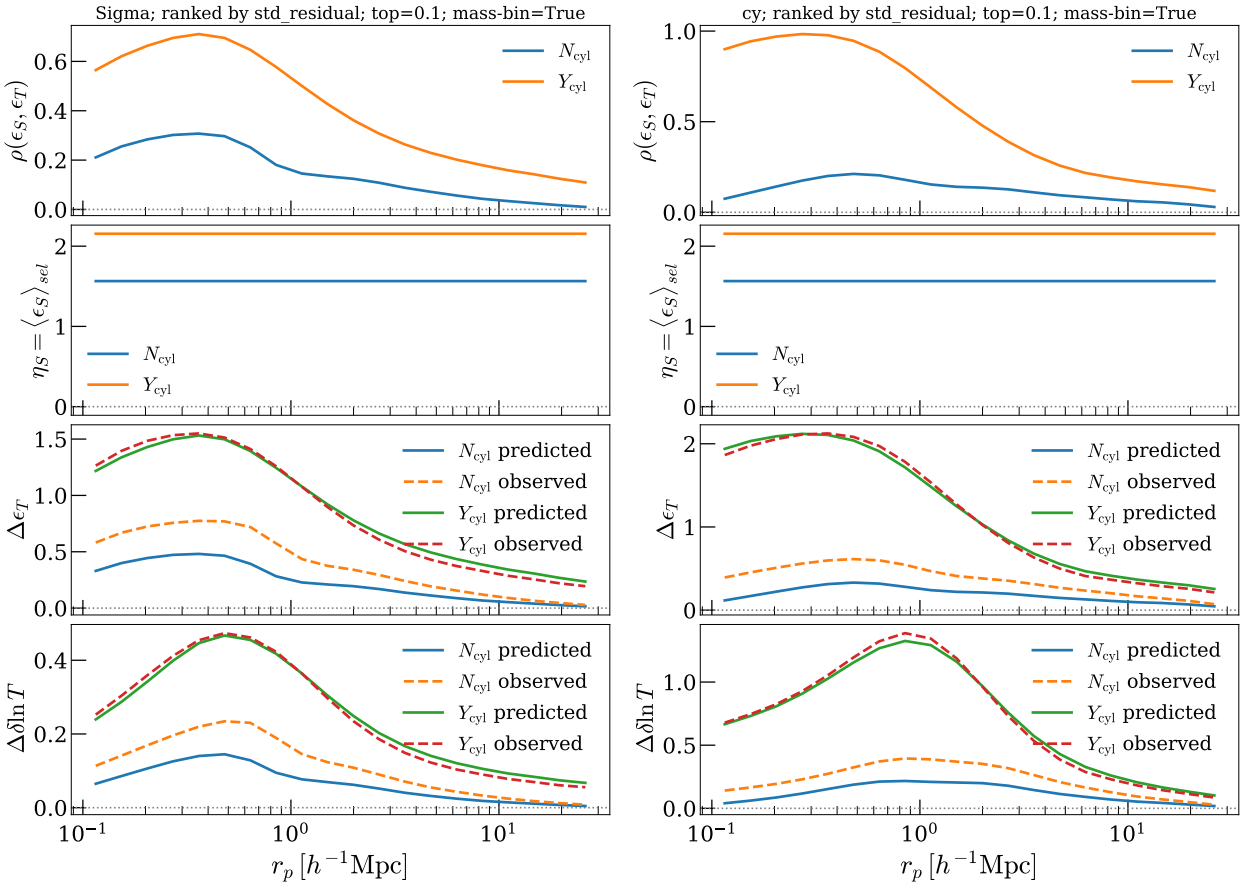
Figure 1: Validation test for the controlled residual-selection response. The left panel shows the result for the lensing surface-density profile $\Sigma(r_p)$, and the right panel shows the result for the Compton-y profile. From top to bottom, each panel shows the correlation coefficient $\rho_{S,T}$, the selection strength η_S, the predicted and observed normalized residual shift $\Delta\epsilon_T$ in Eqs. (11) and (12), and the raw log-residual shift in Eq. (7). I use the ϵ_N and ϵ_Y to rank the halos and selected the top 10% halo without the mass-bin selection.

With mass-bin selection, the mass distributions are nearly matched:

$$\Delta\langle \ln M\rangle_N \simeq +0.0015, \qquad \Delta\langle \ln M\rangle_Y \simeq +0.0007. \tag{26}$$

The mass-trend shift is then close to zero, and the total log shift is almost equal to the residual log shift. The fact that the mass-bin and non-mass-bin tests give similar residual responses shows that **the main result of this controlled test is not driven by the selected mass distribution.**

The simple Gaussian prediction in Eq. (11) captures the qualitative ordering of the selections. It works better for $Y_{\rm cyl}$ than for $N_{\rm cyl}$. For richness, the observed residual shift is larger than the simple prediction, suggesting that the richness residuals and their correlation with the target profiles are not perfectly described by a Gaussian linear model. Nevertheless, the ordering remains the same: $Y_{\rm cyl}$-selection gives the larger residual response.



(a) $\Sigma(r_p)$

(b) Compton-y

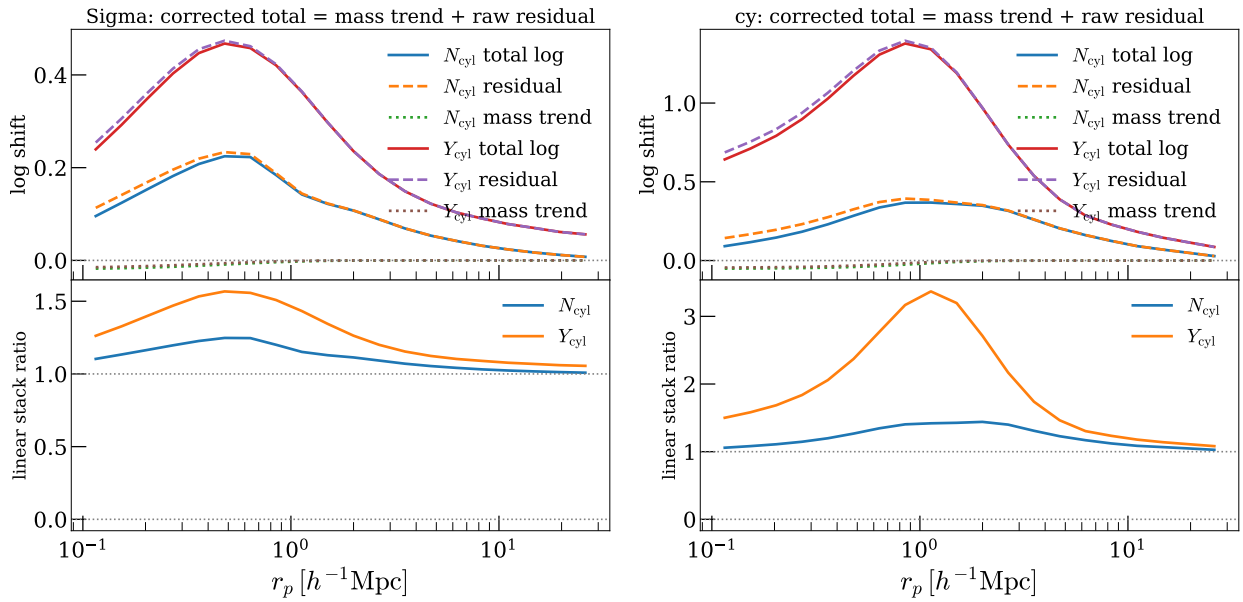
Figure 2: The same as Fig. 1 but use the ϵ_N and ϵ_Y-ranked halos and selected the top 10% halo with the mass-bin selection.

1.6 Conclusion

This validation test does not reproduce the original apparent inconsistency. On the same richness-matched parent halo sample, and with a clean top 10% selection by standardized fixed-mass residual, the larger $Y_{\rm cyl}$ profile residual correlation does lead to a larger $Y_{\rm cyl}$ selection response.

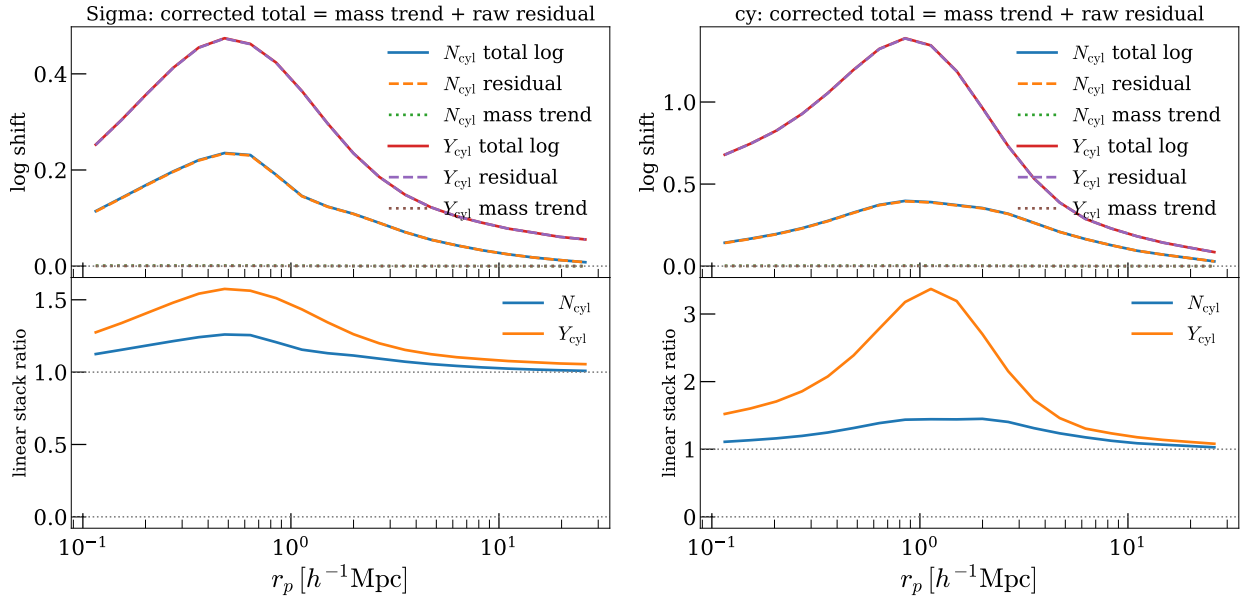
Therefore, if we observe a larger bias for the $N_{\rm cyl}$-selected sample than for the $Y_{\rm cyl}$-selected sample, that difference is unlikely to be caused by the residual correlation coefficient alone. It is more likely caused by differences between the controlled test and the measurement in different cases, such as selection by proxy value rather than residual of proxy, different abundance thresholds, different parent samples, different projection depths, or the use of a linear stacked-signal ratio rather than a mean log-residual shift.

Figures 5 and 6 repeat the same validation test, but now the halos are ranked by the proxy values, $N_{\rm cyl}$ and $Y_{\rm cyl}$, rather than by their standardized residuals. This is closer to the way the selection bias measurements/ratios are performed. The no-mass-bin case (`mass-bin=Flase` in the title) is especially informative. In that case, $N_{\rm cyl}$ selection gives a larger selection strength, $\eta_N > \eta_Y$, and a larger observed target-residual response, even though the residual correlation of the parent sample satisfies $\rho_{Y,T} > \rho_{N,T}$. **This demonstrates that the apparent inconsistency is not caused by the correlation coefficient calculation**



(a) $\Sigma(r_p)$ (b) Compton-y

Figure 3: Test results in Section 1.3 without mass-bin selection.

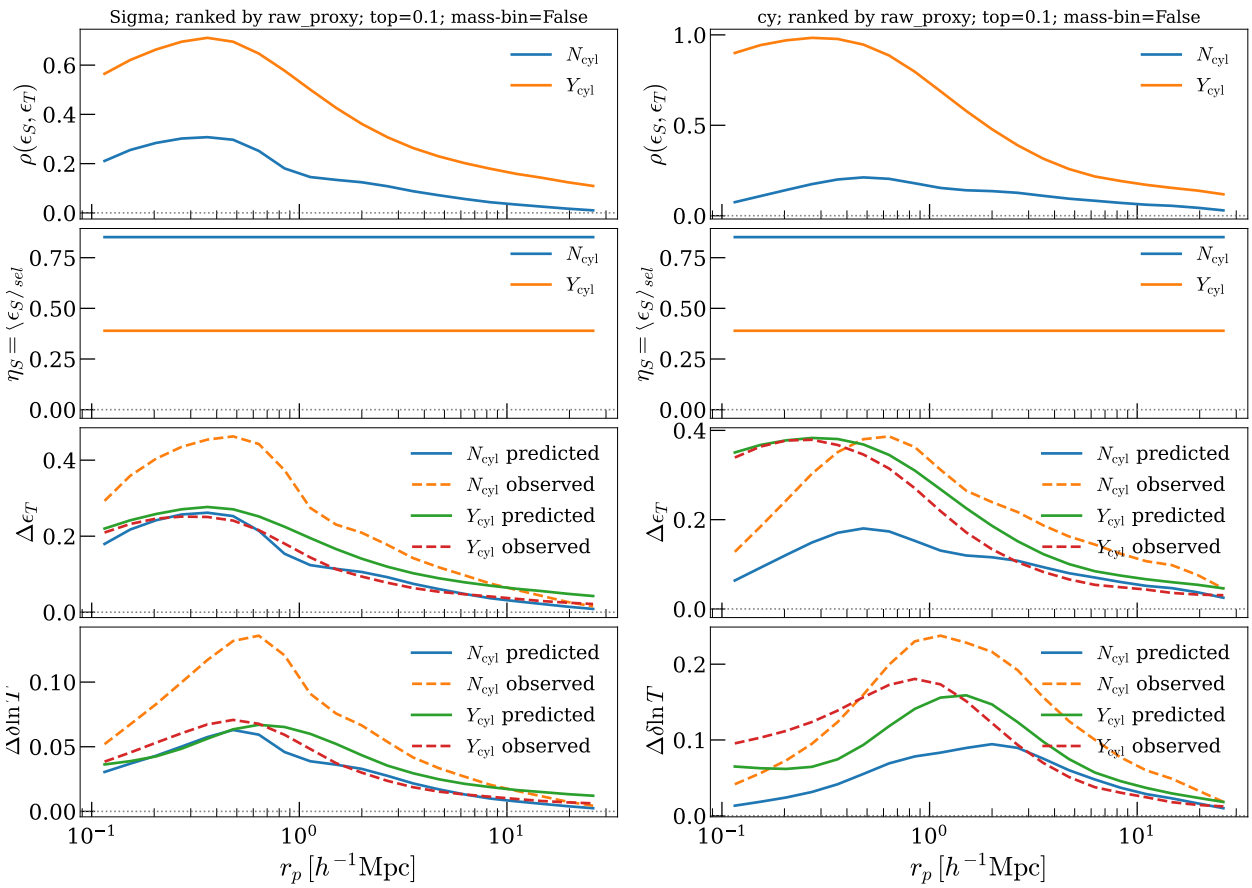


(a) $\Sigma(r_p)$ (b) Compton-y

Figure 4: Test results in Section 1.3 with mass-bin selection.

itself. Instead, it arises because ranking by the proxy changes the selected residual tail. $Y_{\rm cyl}$ selection is more strongly driven by halo mass, while $N_{\rm cyl}$ selection is more sensitive to projected galaxy-richness fluctuations at fixed mass. After comparing to a mass-PDF-weighted expectation, this fixed-mass projection tail can leave a larger residual stacked signal bias for the $N_{\rm cyl}$-selected sample.

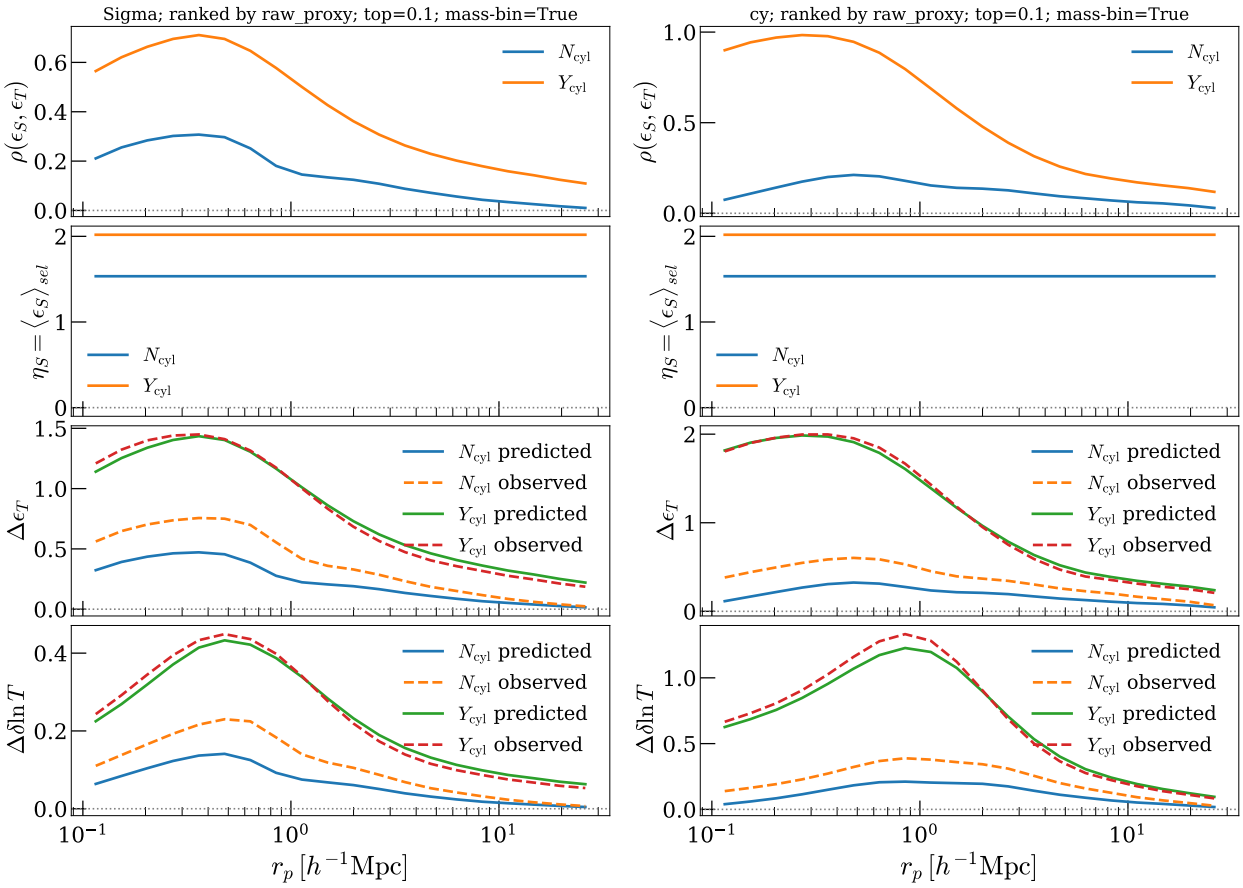
Figures 7 and 8 show the corresponding decomposition into the total mean log shift, the mass trend shift, the log residual shift, and the linear stacked signal ratio. The mass bin selection test removes most of the mass distribution difference, and then the result becomes closer to the clean residual-selection test. Therefore, the discrepancy or the observed "inconsistency" is mainly associated with full sample proxy value ($N_{\rm cyl}$) selection, together with the non-Gaussian/projection-dominated tail of the richness $N_{\rm cyl}$ and the difference between mean log shifts and linear stacked ratios.



(a) $\Sigma(r_p)$

(b) Compton-y

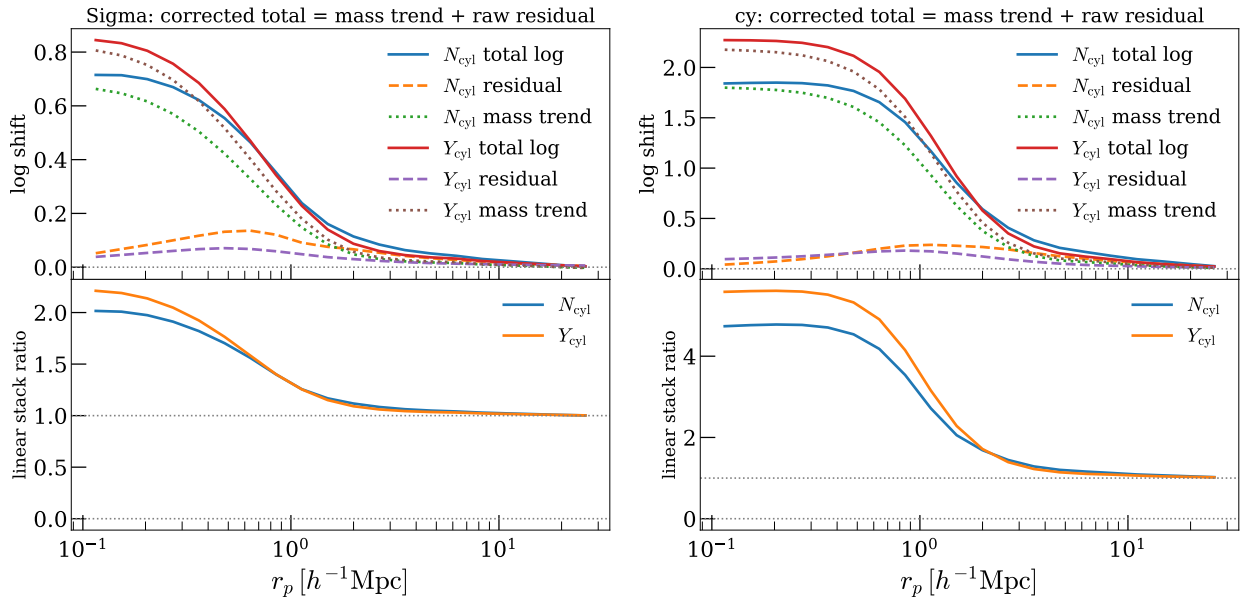
Figure 5: The same as Fig 1 but ranked by $N_{\rm cyl}$ and $Y_{\rm cyl}$, rather than the normalized residual ϵ.



(a) $\Sigma(r_p)$

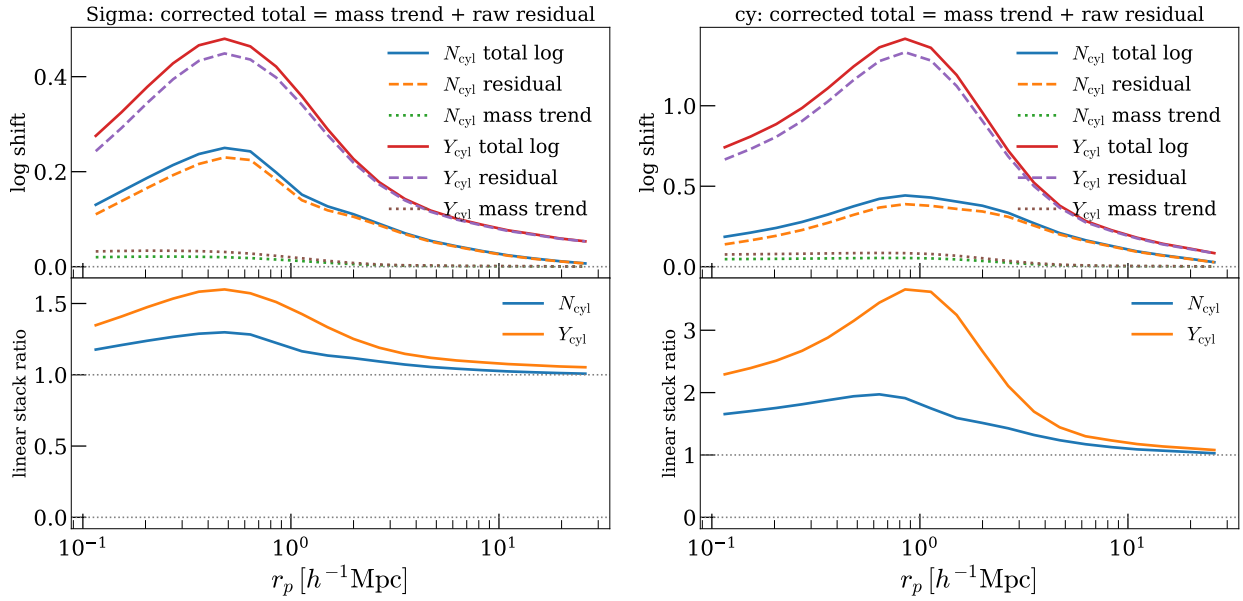
(b) Compton-y

Figure 6: The same as Fig. 1 but use the $N_{\rm cyl}$ and $Y_{\rm cyl}$-ranked halos and selected the top 10% halo with the mass-bin selection.



(a) $\Sigma(r_p)$ (b) Compton-y

Figure 7: Same as Fig3 but ranked by proxy value $N_{\rm cyl}$ $Y_{\rm cyl}$ and without mass-bin selection.



(a) $\Sigma(r_p)$ (b) Compton-y

Figure 8: Same as Fig4 but ranked by proxy value $N_{\rm cyl}$ and $Y_{\rm cyl}$ and with mass-bin selection.